BAHRIA UNIVERSITY (Karachi Campus)

Department of Computer Science

ASSIGNMENT 03 **SPRING 2026**

Course Title: Software Engineering
Course Code: SEN-220
Semester: 4 (A and B)

Start Date: 4 May 2026
End Date: 22 May 2026
Max. Marks: 10

Course Instructor: Hadiqua Fazal

Instructions:

a) This is group assignment; Group size is 3 members in each group
b) Any assignment submitted after the cutoff time will receive zero.
c) Copied /plagiarized assignment will be marked zero

Following characteristics of Complex Computing Problem are focused in this course.

Attribute	Complex Activities
Domain of Application	Software engineering and project management in a real-life software house environment, focusing on the development and deployment of a web-based information system.
Range of resources	Various UML diagram designing tools like Microsoft Visio and Microsoft Project is used for the Illustration of proposed system.
Depth of Knowledge Required	Theoretical and conceptual fundamentals knowledge would be needed to solve this well-defined computing problem.
Depth of Analysis	Students will conduct detailed analysis of requirements, actors, scenarios, and data flows, supported by multiple UML diagrams (Use Case, DFD, State Transition, Network) and project management artifacts (Gantt and CPM charts) to fully understand and refine the proposed system.
Level of interaction	It's based on diagram of UML (Unified Modeling Language) with the purpose of visually representing a system along with its main actors, roles, actions, artifacts or classes, in order to better understand, alter, maintain, or document information about the proposed system.
Innovation	Students will propose and justify alternative design, deployment, and project management strategies (e.g., different architectures, technology stacks, and scheduling approaches) and select an optimized solution based on cost, scalability, and maintainability.
Consequences to society and the environment	By applying the designing, scheduling and costing tools on the prototype. It will reduce the implementation errors in real scenarios.

Project Overview and Foundation

This project requires your team to ***analyze*** the complete scope of a real-life software house environment. You'll gain exposure to professional roles and the full Software Development Life Cycle (SDLC), starting with detailed requirements analysis. The initial focus is on design: you must create detailed Wireframes and UI/UX Mockups to visualize and define the user interface before starting any implementation.

System Development and Deployment

The primary goal is to transform your design plans into a fully functional software system. This requires developing and integrating all system containers: the Frontend, the Backend, and a complete Database. Each module must be implemented and tested, including unit testing, integration testing, and basic user acceptance testing, with test cases and results properly documented. The final application must be deployed on a hosting platform. Throughout development, all source code must be rigorously managed using GitHub for version control.

Tools and Technologies

Frontend:

HTML, CSS, JavaScript (with or without frameworks like Bootstrap, React, Vue, etc.)

Backend:

Any one stack, for example:

- Node.js + Express
- Django / Flask (Python)
- Laravel / PHP
- ASP.NET / Java Spring, etc.

Database:

MySQL, PostgreSQL, SQLite, MongoDB, or similar.

Implementation and testing of the modules/project

Each group will implement the core modules and integrate them into a working system. Basic tests (unit and integration) must be performed, and main test cases, results, and fixes should be briefly documented.

Required Documentation and Project Control

To ensure a robust understanding of the system's architecture and process, you must submit comprehensive documentation. This includes several analysis diagrams to model the system's structure and behavior, and two key charts for project management. The documents must cover the overall System Context and data flow.

List of Required Deliverables

1. Design Artifacts: Detailed Wireframes and UI/UX Mockups.
2. Functional System: A deployed, fully functional application (Frontend, Backend, and Database).
3. Code Management: Complete source code repository managed via GitHub.
4. Modeling Diagrams:
   - Use Case Diagram
   - Data Flow Diagram (DFD) up to Level 1 (illustrating system Components)
   - State Transition Diagram
   - Network Diagram (to define System Context and deployment)
   - Risk Registers
5. Project Management Charts:
   - Gantt Chart
   - Critical Path Method (CPM) Chart

Marking Scheme	Marks
Design Artifacts (Wireframes & Mockups)	2
Functional System (Code, Implementation, Integration, Testing, Deployment)	2
Modeling Diagrams (Use Case, DFD, State Transition, Network)	2
Project Management Charts (Gantt & CPM)	1
Project Progress Reports (Participation of Individual team members)	3
Total Marks	10

End of Assignment